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CS23B1013

BUSI classification

Introduction

Breast cancer is one of the most common cancers affecting women worldwide. Early detection plays a crucial role in improving survival rates and treatment outcomes. Medical imaging techniques such as ultrasound are widely used for detecting abnormalities in breast tissue because they are non-invasive, safe, and cost-effective.

However, analyzing ultrasound images manually can be time-consuming and depends heavily on the expertise of radiologists. Recent advancements in deep learning and computer vision allow automated systems to assist doctors in identifying abnormal patterns in medical images.

In this project, a Convolutional Neural Network (CNN) model is developed to classify breast ultrasound images into three categories:

- Benign (B) – non-cancerous tumor
- Malignant (M) – cancerous tumor
- Normal (N) – healthy breast tissue

The goal of this project is to train a deep learning model that can accurately classify ultrasound images and evaluate its performance using standard metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC curve.

Dataset Description

The dataset used in this project is the **BUSI (Breast Ultrasound Images Dataset)**.

The dataset contains ultrasound images of breast tissue collected from patients with different medical conditions.

Class	Description	Number of Images
Benign	Non-cancerous tumors	**437 images**
Malignant	Cancerous tumors	**210 images**
Normal	Healthy breast tissue	**133 images**

Total Images

780 ultrasound images

Image Properties

- Image type: Ultrasound medical images
- Format: PNG
- Color type: Grayscale
- Resolution: varies (resized to **224×224** for training)

Methodology

The following steps were used in the project:

1. Data Preprocessing

The dataset images were preprocessed before training the model.

Preprocessing steps include:

- Resizing images to **224 × 224 pixels**
- Converting grayscale images to **3 channels**
- Normalizing image pixel values
- Converting images to tensors for PyTorch

2. Dataset Splitting

The dataset was divided into:

Dataset	Percentage
Training set	80%
Validation/Test set	20%

Model Selection and Training

A **Convolutional Neural Network (CNN)** was used for the classification task.

CNNs are widely used in image classification because they automatically learn spatial features such as edges, textures, and patterns.

Model Architecture

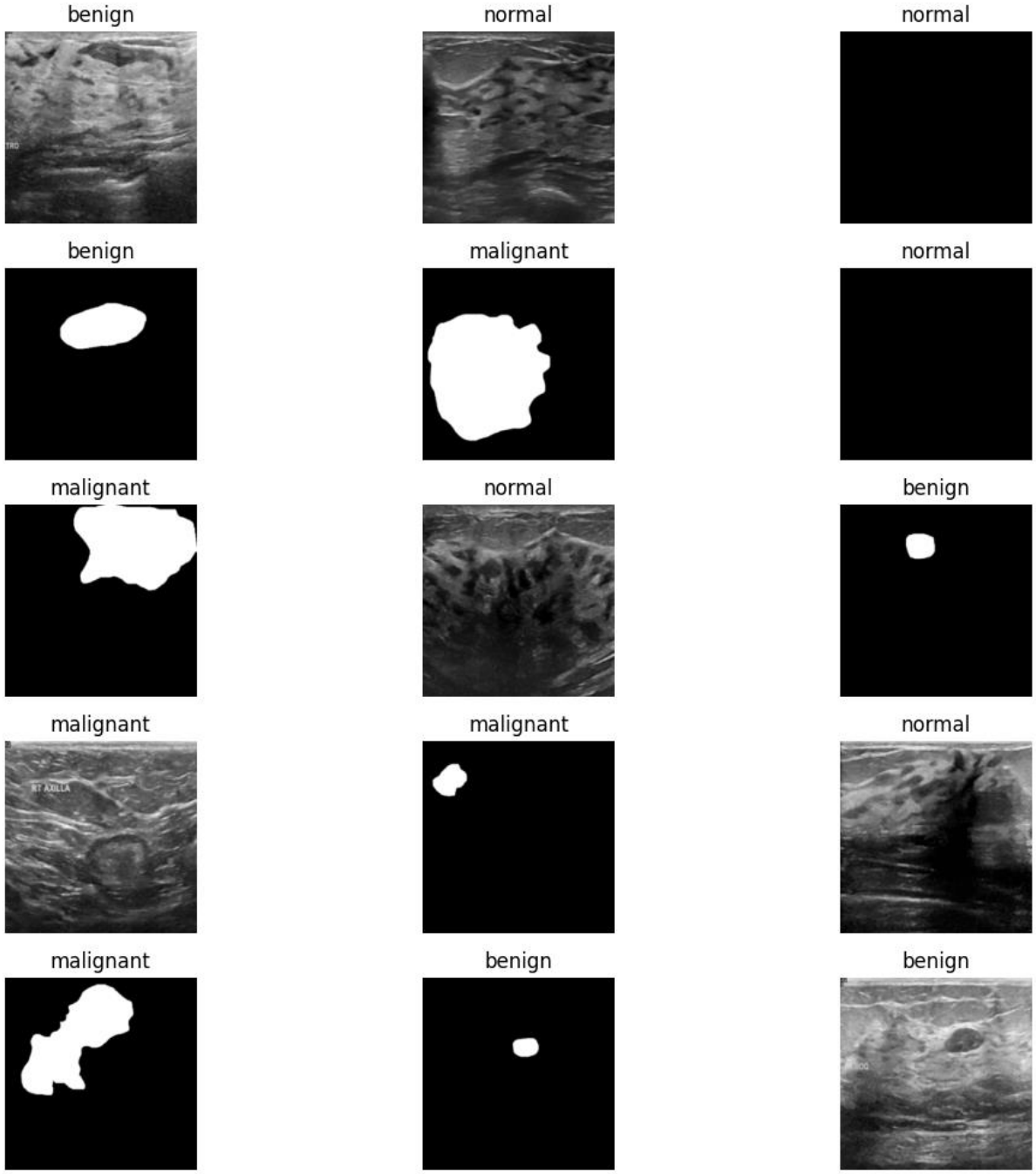
The CNN model contains the following layers:

1. **Convolution Layer**
2. **ReLU Activation**
3. **Max Pooling**
4. **Fully Connected Layers**
5. **Softmax Output Layer**

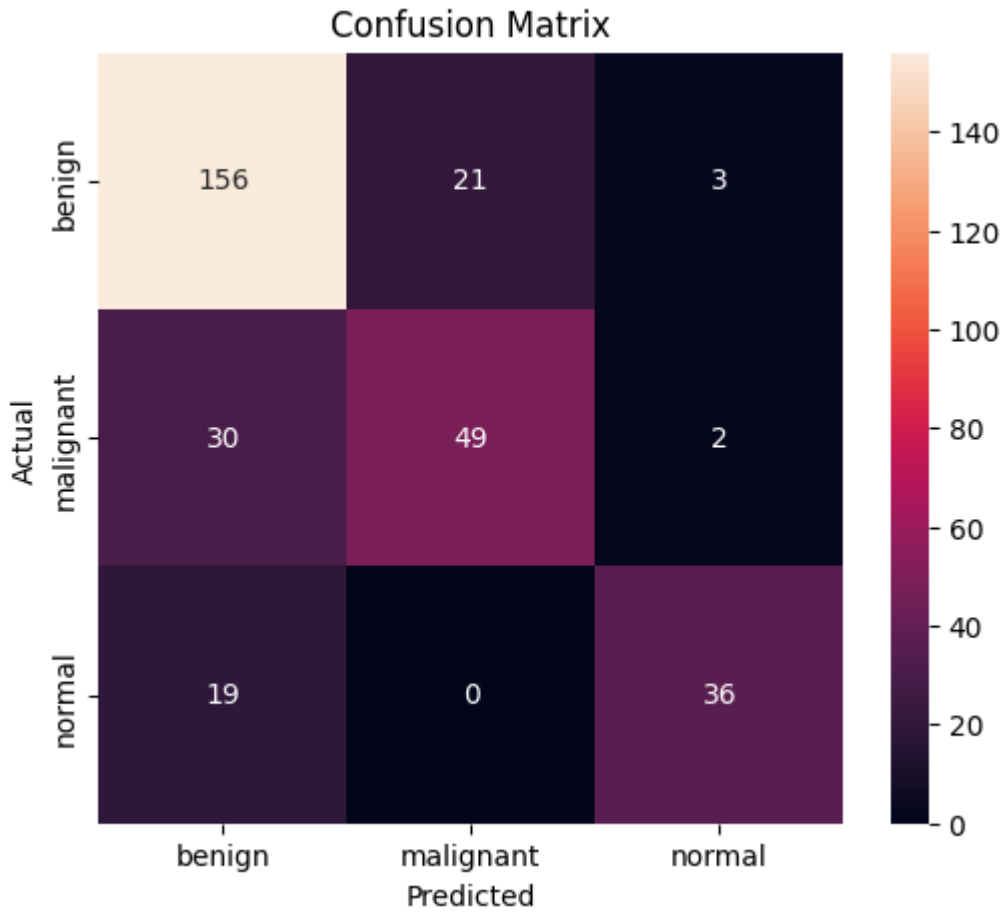
The model outputs probabilities for the three classes

Parameter	**Value**
Optimizer	Adam
Loss Function	CrossEntropyLoss
Batch Size	32
Epochs	10
Image Size	224 × 224

Sample Dataset Images



Evaluation Metrics

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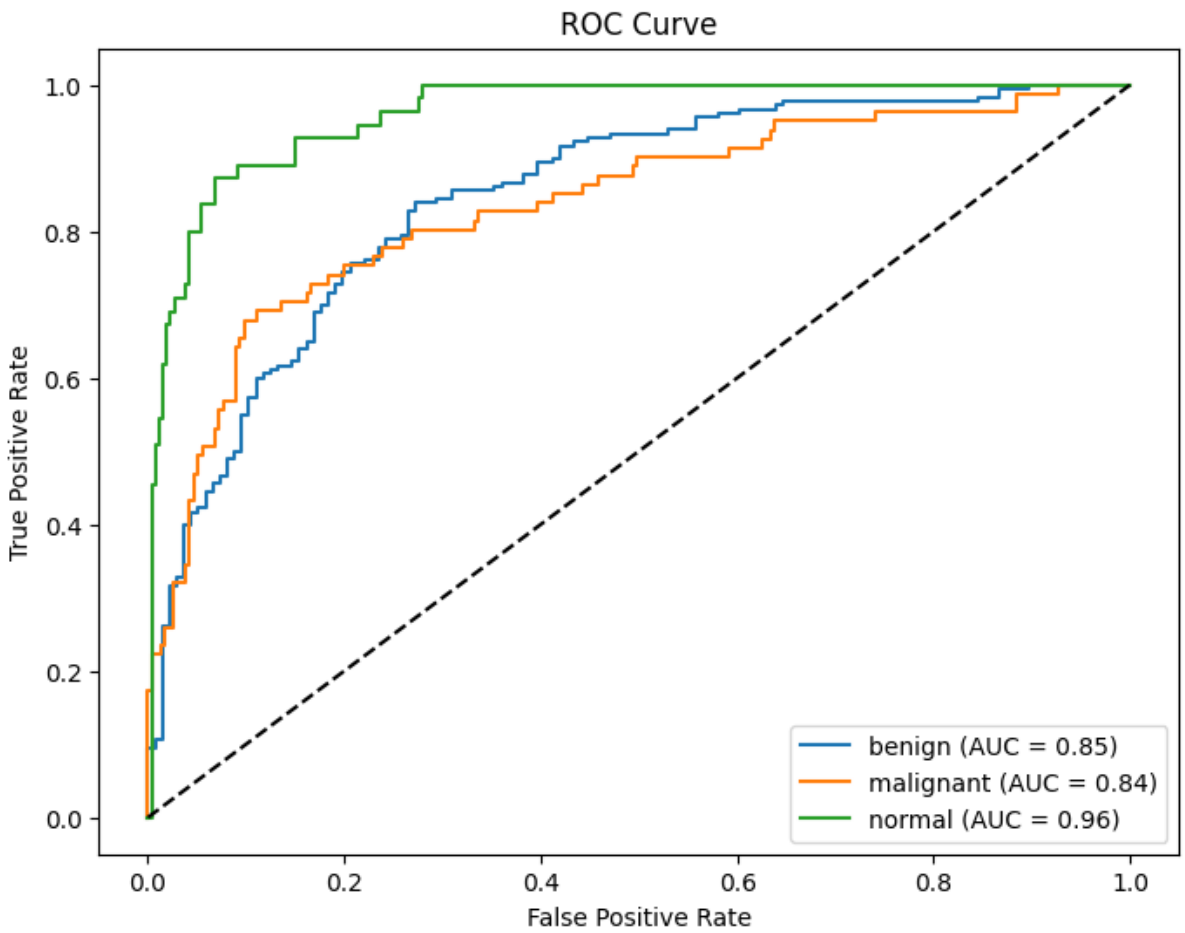
Classification Report

              precision    recall  f1-score   support

      benign       0.76      0.87      0.81       180
   malignant       0.70      0.60      0.65        81
      normal       0.88      0.65      0.75        55

    accuracy                           0.76       316
   macro avg       0.78      0.71      0.74       316
weighted avg       0.77      0.76      0.76       316
```



Conclusion

In this project, a deep learning model was developed to classify breast ultrasound images from the BUSI dataset into three categories: benign, malignant, and normal. A Convolutional Neural Network (CNN) was trained on the dataset after preprocessing and splitting it into training and validation sets. The model's performance was evaluated using metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. The results show that deep learning can effectively assist in automatic breast cancer detection, helping support medical diagnosis and early detection.